

Worldwide IT security problems - viruses, hackers and global cyber threats

Global IT Security threats and viruses

1. What are worldwide IT security problems?

Definition:

IT security problems are global issues that involve the **unauthorized access, damage, or misuse of digital systems**, affecting individuals, businesses and governments.

Examples of global threats:

- data breaches
- cyberattacks on hospitals, airports, or banks
- spread of viruses across continents
- ransomware locking national infrastructure

Why this matters globally?

- cyber threats don't respect borders
- attackers can strike from anywhere
- international cooperation is needed

2. Computer viruses and malware

What is a virus?

A type of **malware** (malicious software) that **spreads between files or systems**, often replicating itself and damaging data.

Other types of malwares:

Type	Description
Trojan	Pretends to be useful software but is actually harmful. They are waiting to be launched by a computer user's click or by another program that calls it.
Worm	Spreads itself without user action, often over a network
Spyware	Secretly collects user data
Ransomware	Locks files or systems until a ransom is paid
Adware	Displays unwanted ads, sometimes redirects web traffic
Rootkits	Hide the existence of other malware

Famous virus examples:

- ILOVEYOU (2000): Spread via email, caused billions in damages
- WannaCry (2017): Ransomware that hit 200,000+ systems in 150 countries
- Stuxnet (2010): Targeted Iran's nuclear facilities, believed to be state-sponsored

3. How viruses spread?

- infected email attachments
- fake downloads (e.g., cracked software)
- USB drives
- malicious websites
- network vulnerabilities

1. Email attachments

A very common method. A virus hides inside an attachment (e.g., .exe, .docm, .xlsm, .zip). When the user opens it, the virus runs and infects the computer.

Example: Fake invoice or delivery notice with a malicious attachment.

2. Malicious downloads

Viruses can be bundled with pirated software, cracked programs, fake updates, or infected apps.

Example: Downloading a fake “Chrome update” from an untrusted site.

3. File sharing (USB, external drives, shared folders)

Viruses can copy themselves to USB drives or network shares. When the drive is plugged into another computer, the virus executes.

Example: “autorun” viruses that used USB drives to infect multiple machines.

4. Exploiting software vulnerabilities

Some viruses spread automatically by exploiting security flaws (no user clicks required).

Example: Worms like WannaCry exploited Windows SMB vulnerabilities to spread across networks.

5. Infected websites (drive-by downloads)

Just visiting a compromised website can trigger a silent infection if the browser or plugins are vulnerable.

Example: Malicious JavaScript that forces the browser to run harmful code.

6. Malicious macros in Office documents

Office documents with macros (.docm, .xlsm) can run harmful code once macros are enabled. Macro - set of recorded or written instructions that automates repetitive tasks by executing multiple actions with a single command.

Example: “Enable macros to view content” trick used in phishing emails.

7. Instant messaging and social media links

Viruses spread by sending infected links or files through:

- Messenger
- WhatsApp Web
- Facebook messages
- Discord

Example: A message like “Is this your photo?” with a malicious link.

8. Network propagation

On corporate networks, viruses may:

- scan for vulnerable computers
- copy themselves to shared folders
- use stolen credentials to move laterally

Example: Viruses that scan the LAN and infect all reachable machines.

9. Infected software updates (supply chain attacks)

A legitimate software provider is compromised, and **malicious code is inserted into updates.**

Example: The SolarWinds supply chain compromise.

10. Email account hijacking

Once a virus infects one machine, it may steal login credentials and automatically send itself to the victim's contacts.

Example: A worm that emails itself to every contact found in Outlook.

Summary

Computer **viruses spread by:**

- deception (tricking users to run infected files)
- exploitation (running an infected file)
- replication (copying themselves to new hosts)

Hackers and cybercrime

1. Who are hackers?

Definition:

Hackers are individuals or groups who **manipulate computer systems**, often by bypassing security systems.

Types of hackers:

Hacker Type	Goal/Purpose
White Hat	Ethical hackers – test systems to improve security
Black Hat	Malicious hackers – steal data, cause damage
Gray Hat	Breaks rules but without harmful intent
Hacktivists	Hack for political or social causes
Script Kiddies	Use existing tools without real knowledge
State-sponsored	Work for governments to conduct cyber-espionage

Not all hacking is illegal - **ethical hacking** is a respected profession.

Ethical hacking is the **authorized and legal practice of testing and evaluating** the security of computer systems, networks, or applications by **simulating cyberattacks**. It is performed by **trained professionals** - ethical hackers or penetration testers, whose goal is to find and fix security weaknesses before malicious hackers can exploit them.

Real-world examples:

- Anonymous – Hacktivist group targeting governments
- SolarWinds Hack (2020) – Alleged Russian cyberattack on U.S. infrastructure
- Equifax Breach (2017) – Stolen data of 147 million Americans

2. Impact of global cybercrime

- **Financial losses:** Trillions of dollars globally
- **Personal data theft:** Identity fraud, spam, scams
- **National security:** Governments and infrastructure are targeted
- **Public health:** Hospitals hit by ransomware (e.g., during COVID-19)

Interesting fact: Cybercrime is now **more profitable** than the illegal drug trade.

3. How to protect against global threats?

Basic protection tips:

- Use antivirus and keep it updated
- Don't open unknown links or attachments
- Install system and app updates regularly
- Use strong, unique passwords
- Enable two-factor authentication
- Avoid pirated software or unofficial downloads

For organizations:

- Firewalls and intrusion detection systems
- Employee cybersecurity training
- Backup systems and disaster recovery plans
- Security audits and penetration testing

4. The relationship between artificial intelligence and cyber defense

Artificial intelligence (AI) plays a key role in modern cyber defense by helping systems detect, prevent and respond to cyber attacks faster and more accurately than humans alone.

In practice:

- **Threat detection** - AI analyzes huge volumes of network traffic, logs, and user behavior to spot anomalies that may indicate malware, intrusions, or data breaches.
- **Attack prevention** - Machine learning models can recognize known attack patterns (phishing, ransomware, DDoS) and block them in real time.
- **Incident response** - AI helps automate responses - isolating infected devices, prioritizing alerts and suggesting mitigation steps.

- **Adaptation to new threats** - Unlike static rule-based systems, AI can learn from new attacks and evolve as attackers change techniques.

In short, AI strengthens cyber defense by adding **speed, scalability, and adaptability**, which are essential against today's complex and fast-moving cyber threats.

Key terms summary

Term	Definition
Malware	Any software created to harm, steal, or spy on systems
Virus	A self-replicating malware that spreads between files or devices
Trojan	A program that appears useful but is harmful
Hacker	Someone who breaks into computer systems, ethically or maliciously
Ransomware	Malware that locks files and demands payment
Ethical hacker	Security expert who finds and fixes vulnerabilities
Cybercrime	Crime committed using computers or the internet